

Worldwide greenhouse gas emissions of green hydrogen production and transport

Received: 18 August 2023

Accepted: 27 May 2024

Published online: 21 June 2024



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Large-scale introduction of green hydrogen is envisioned to play an important role in reaching net-zero greenhouse gas emissions. The production and transport of green hydrogen itself is, however, not free from emissions. Here we assess the life-cycle greenhouse gas emissions for 1,025 planned green hydrogen facilities, covering different electrolyser technologies and renewable electricity sources in 72 countries. We demonstrate that the current exclusion of life-cycle emissions of renewables, component manufacturing and hydrogen leakage in regulations gives a false impression that green hydrogen can easily meet emission thresholds. Evaluating different hydrogen production configurations, we find median production emissions in the most optimistic configuration of 2.9 kg CO₂ equivalents (CO₂e) kg H₂⁻¹ (0.8–4.6 kg CO₂e kg H₂⁻¹, 95% confidence interval). Including 1,000 km transport via pipeline or liquid hydrogen shipping adds another 1.5 or 1.8 kg CO₂e kg H₂⁻¹, respectively. We conclude that achieving low-emission green hydrogen at scale requires well-chosen production configurations with substantial emission reductions along the supply chain.

Many actors expect hydrogen to become an important element in the transition towards a net-zero greenhouse gas (GHG) emissions society^{1–6}. As a complement to electrification, hydrogen can contribute to the decarbonization of industrial, transport and energy systems⁷. The current, fossil fuel-based hydrogen market of 100 million tonnes per year is projected to expand towards 300–800 Mt yr⁻¹ of green hydrogen produced with electrolysis using renewable electricity by 2050, and up to 200 Mt yr⁻¹ of blue hydrogen produced via steam methane reforming combined with carbon capture and storage⁸. Where renewable electricity and water are sufficiently abundant, green hydrogen might be produced, converted and exported to countries where production is more expensive or constrained, eventually turning hydrogen and its derivatives into internationally traded commodities⁹. Domestic hydrogen strategies, such as those of the European Union (EU) and Japan, foresee

a large and increasing role for hydrogen imports in reaching emission reduction targets, while other countries, such as Australia, Mauritania and Chile, position themselves to become hydrogen exporters¹.

To quantify GHG emissions of hydrogen production, the International Energy Agency (IEA) and most proposed regulations and certification systems follow a methodology that assigns zero emissions to renewable electricity use and excludes emissions from equipment manufacturing, yielding a footprint of 0 kg CO₂ equivalents (CO₂e) kg H₂⁻¹ (ref. 10). However, from a life-cycle perspective, hydrogen production emissions are far from zero and depend on the electricity generation technology^{11,12}, electrolyser technology¹³ and hydrogen production configuration (that is, relative sizing of the different components and the type of grid integration)¹⁴. Additional emissions arise from transporting hydrogen via pipeline¹⁵ or ship^{16,17}, and for conversion

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into derivatives that are more easily shipped, like ammonia¹⁸. As it is generally assumed that global climate targets are jeopardized if rapid investments in green hydrogen supply chains are not realized^{19–21}, clarity is urgently needed on how these different steps add up for different technological and geographical contexts, and what combination of factors would lead green hydrogen projects across the world to result in low emissions.

Here, we quantify GHG footprints of green hydrogen, considering the full production, conversion and transport chain. Since emissions will depend on local circumstances, we analyse GHG footprints for 1,025 individual green hydrogen facilities worldwide planned in 2030, covering different electrolyser technologies and electricity sources in 72 countries²². We show that the GHG footprint considerably varies between electricity sources and project configurations and find that, from a full life-cycle perspective, only a limited share of the potential green hydrogen production would be consistent with emission thresholds, such as the EU sustainability taxonomy (21–85%) or the US Clean H₂ Standard (23–95%). We further found that transport of green hydrogen to the end user can greatly increase the GHG footprint, depending on transport mode and distance.

GHG emissions of green hydrogen production

We calculate project-specific hydrogen cradle-to-production gate GHG emissions based on electrolyser-specific water, energy and material requirements, and electricity source documented in the IEA Hydrogen Projects Database²². We use country-specific renewable electricity GHG footprints based on meteorological conditions^{23,24}, facility type-specific hydropower footprints²⁵ and national 2030 grid mixes modelled for a 2 °C policy scenario²⁶. As many of the planned projects are located in water-stressed regions⁸, we assume that sea water desalination is required proportional to the local level of water stress²⁷ (excluding landlocked countries).

Most of the studied hydrogen projects are stated to rely on renewable electricity generation capacity, which we assume is additional and would not have been built without the electrolyser. For these projects, to account specifically for the intermittency of solar and wind energy generation, we consider three hydrogen production configurations with different relative sizing of components and grid integration (Table 1). These configurations are based on the cost-optimal size ratio of renewable, battery and electrolyser capacity as identified in previous studies^{14,28}. Since electrolysers' high investment costs prohibit excessive oversizing to accommodate peaks in renewable energy output, the electrolyser utilization factor is typically between 0.4 and 0.6 when running exclusively on renewable energy^{14,28}, depending on local circumstances, with a central value of 0.5 (this means that annual hydrogen production is half of continuous production at full capacity).

In the first, 'off-grid: curtailment' configuration, there is no grid connection, the renewable capacity is enlarged to double the size of the installed electrolyser capacity and a lithium-ion battery is installed. Additional electricity generation beyond electrolyser and battery capacity is curtailed, and emissions associated with curtailed electricity are allocated to hydrogen. In the 'grid-connected: power export' configuration, additional electricity produced in the otherwise same configuration can be used by other consumers via a grid connection. This scenario could be applicable where renewable generation substitutes flexible fossil power but may not be applicable in areas where renewable peak production regularly meets demand²⁹. In the final 'grid-connected: power import' configuration, renewable peak capacity matches electrolyser capacity such that no batteries are needed, and no surplus electricity can be produced. Instead, grid electricity is imported to top off intermittent renewables and to realize an electrolyser utilization factor of 1.

No separate configurations are distinguished (configuration independent) for hydrogen projects that are planned without newly built electricity generation capacity (12% in database), which will instead

Table 1 | Green hydrogen production configurations based on intermittent renewables

	Off-grid: curtailment	Grid-connected: power export	Grid-connected: power import
Electrolyser utilization factor	0.5	0.5	1
Grid connection	No	Yes	Yes
Renewable peak capacity compared with electrolyser capacity	Double	Double	Equal
Battery	Yes	Yes	No
Curtailment factor	0.5	0	NA

'Off-grid: curtailment' with curtailment of surplus electricity; 'grid-connected: power export' with a buyer of surplus electricity; 'grid-connected: power import' in a hybrid system where renewables are topped off with grid electricity, based on the configurations in Palmer et al.²⁸ and Terlouw et al.¹⁴. The default electrolyser utilization factor is 0.5 in the first two configurations (when relying on renewables) and is varied between 0.4 and 1 in the sensitivity analysis.

rely either on continuous sources like nuclear or hydroelectricity (electrolyser utilization factor 1), on a continuous grid connection (electrolyser utilization factor 1) or on excess renewable electricity. Due to the limited projected availability of excess electricity in most grid systems in 2030^{30,31}, the latter is represented by an electrolyser utilization factor of 0.05.

Electricity generation to power the electrolyser is typically the largest contributor to production emissions (Fig. 1), except when excess electricity is used, where emissions from electricity use are considered zero^{32,33}. Life-cycle GHG emissions of solar photovoltaic electricity exceed those of wind electricity owing to higher emissions associated with manufacturing of solar panels compared with wind turbines. GHG emissions from heat generation, water desalination and electrolyser and battery manufacturing typically constitute a much smaller part of total production emissions. These manufacturing emissions are distributed over the total amount of hydrogen produced over the electrolyser lifetime, meaning they are about twice as large when running on intermittent renewables (Fig. 1a,b) compared with continuous hydrogen production (Fig. 1c,d). When running on excess electricity, the lower utilization factor results in even higher emissions from electrolyser manufacturing (Fig. 1d).

GHG emissions of green hydrogen production are between 0.3 and 36.5 kgCO₂e kg H₂⁻¹ across planned projects, depending on the hydrogen production configuration and electricity source (Fig. 2). In the 'off-grid: curtailment' configuration, overproduced power is not valorized (Fig. 2a), resulting in higher emissions than in the 'grid-connected: power export' configuration, where this surplus is valorized and associated emissions are not allocated to hydrogen (Fig. 2b). Emissions in the 'grid-connected: power import' configuration are highest, due to the use of grid electricity to achieve continuous operation. Here, production emissions in 71% of the onshore wind and 88% of the solar projects exceed those of grey hydrogen, while offshore wind has lower emissions due to its higher capacity factor (Fig. 2c).

In our estimate, median production emissions across all 1,025 planned electrolytic hydrogen projects are 2.9 kgCO₂e kg H₂⁻¹, when considering the most favourable 'power export' configuration for intermittent renewables. The variability around this median is considerable, with emissions ranging from 0.8 kgCO₂e kg H₂⁻¹ at the low end to 4.6 kgCO₂e kg H₂⁻¹ at the high end, representing the 95% interval. Across the board, however, only hydrogen production from wind electricity (in the 'curtailment' and 'power export' configurations), hydropower, nuclear energy and excess renewables has emissions below the emission thresholds set for clean hydrogen by the USA (4.0 kgCO₂e kg H₂⁻¹), EU (3.0 kgCO₂e kg H₂⁻¹) and UK (2.4 kgCO₂e kg H₂⁻¹)¹⁰. Hydrogen produced from solar only partly stays below the US threshold, and

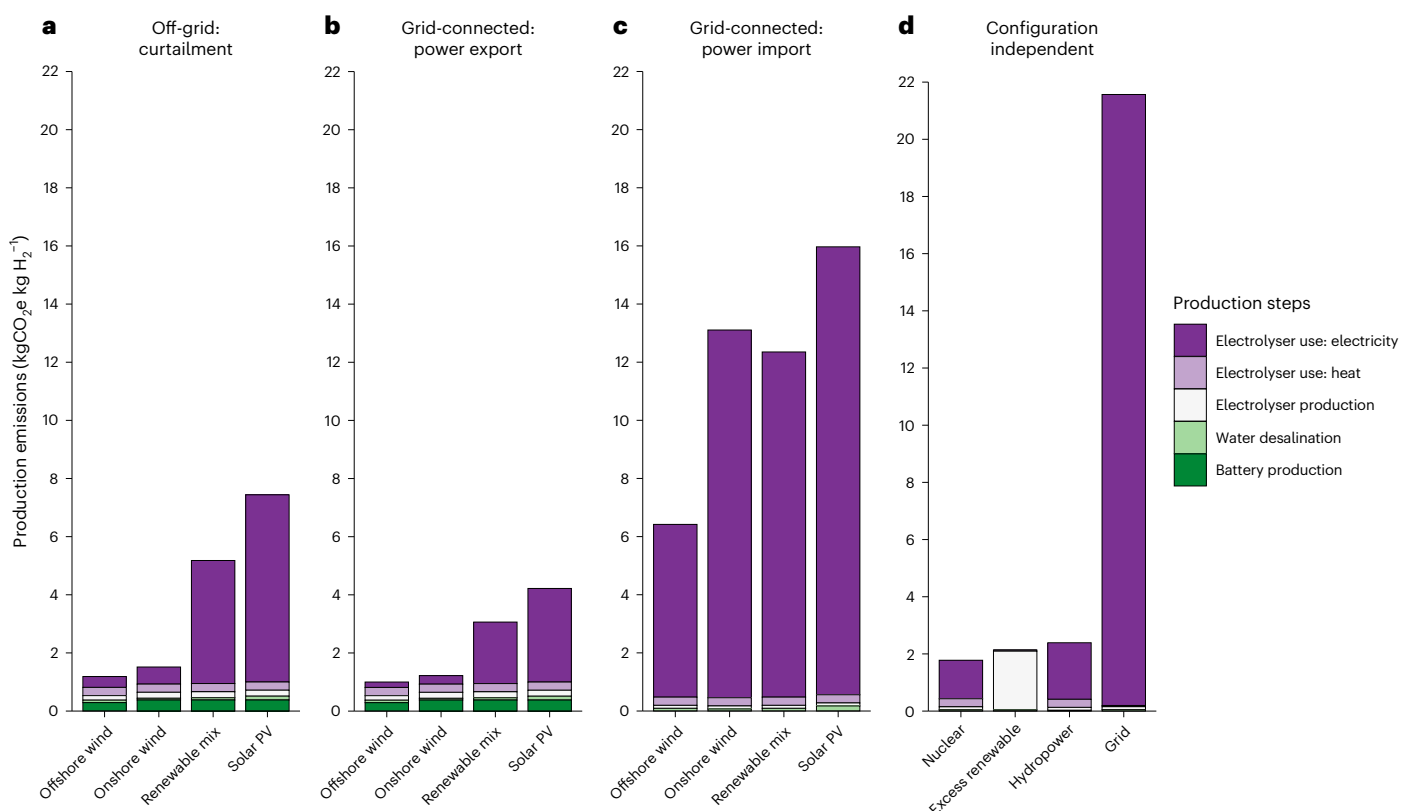


Fig. 1 | Contribution of production steps to the overall GHG emissions of green hydrogen produced from various electricity sources.

a–d. Contributions are medians across the green hydrogen projects by each intermittent renewable electricity source in the ‘off-grid: curtailment’

configuration (a), ‘grid-connected: power export’ configuration (b) and ‘grid-connected: power import’ configuration (c) and for all configuration-independent electricity sources (d).

only in the ‘power export’ configuration. For comparison, blue hydrogen produced from steam methane reforming could result in 2.3–8.9 kgCO₂e kg H₂⁻¹, assuming a CO₂ capture rate of 93% can be achieved (the range reflects assumptions on upstream methane leakage between 0.2% and 8%)³⁴. Overall, our results on green hydrogen production imply that power source and hydrogen production configuration have to be carefully selected, and that further emission reductions in the production process are necessary, to consistently deliver truly green hydrogen.

Variation in emissions across hydrogen projects with the same renewable electricity source largely results from differences in location and the corresponding footprint of the renewable electricity (Fig. 3 and Supplementary Fig. 2). Emissions of renewable electricity are lowest in locations with high solar irradiation or wind speeds due to larger total electricity production over the lifetime of the energy facilities and, therefore, lower emissions per kilowatt hour^{23,24}. Hydrogen production emissions using wind electricity are below 3.0 kgCO₂e kg H₂⁻¹ in the ‘power export’ configuration in most locations, except close to the Equator (Fig. 3b). Production emissions using solar electricity only stay below this threshold very locally in South America (Fig. 3e), and median solar-based production emissions are 36% higher in Europe compared with South America (Supplementary Fig. 2). Emissions in the ‘power import’ configuration depend on local capacity factors of the renewable sources and the modelled GHG intensity of national grid mixes in 2030 (ref. 26).

Hydrogen production emissions show little variation with regard to the uncertainty and variability that have been identified in previous work for the various production parameters (Supplementary Information section 8). These include equipment lifetime (5–8% change in median GHG emissions for the high to low end of the literature range),

electrolyser efficiency (5% change for conservative and progressive electricity requirement estimates), number of cooling water cycles (10% increase if cooling water is not recirculated), assumed required battery capacity (9% change in emissions based on a 66% change in battery capacity) and site optimization within the specified country (5–6% change based on using the 90th percentile or median irradiation and wind speed instead of the 75th percentile). Varying the electrolyser utilization factor when running exclusively on renewables from 0.5 to the literature-based range of 0.4–0.6 results in a 3–5% change in emissions. Further increasing the electrolyser utilization factor to 1 would result in a limited emissions decrease of 10% (‘power export’) or 6% (‘curtailment’), as only allocated emissions of equipment manufacturing are affected (that is, allocation over the amount of hydrogen produced over the lifetime). In contrast, hydrogen production emissions from excess renewables are sensitive to the assumptions made: if its electricity generation emissions are attributed to hydrogen, emissions approximately double (Supplementary Fig. 3).

GHG emissions of green hydrogen conversion and transport

As green hydrogen is expected to become an internationally traded commodity, its life-cycle emissions should also include emissions from its conversion and transport. In our analysis of hydrogen transport, we include all documented projects²², except those powered solely by grid electricity, and we use the ‘power export’ configuration for intermittent renewables. We consider the most technologically mature and cost-efficient hydrogen transport options²: (1) pipeline, (2) shipping of liquefied hydrogen followed by regasification at the destination, and (3) conversion to ammonia and shipping, with reconversion (‘cracking’) to hydrogen at the destination (Fig. 4a).

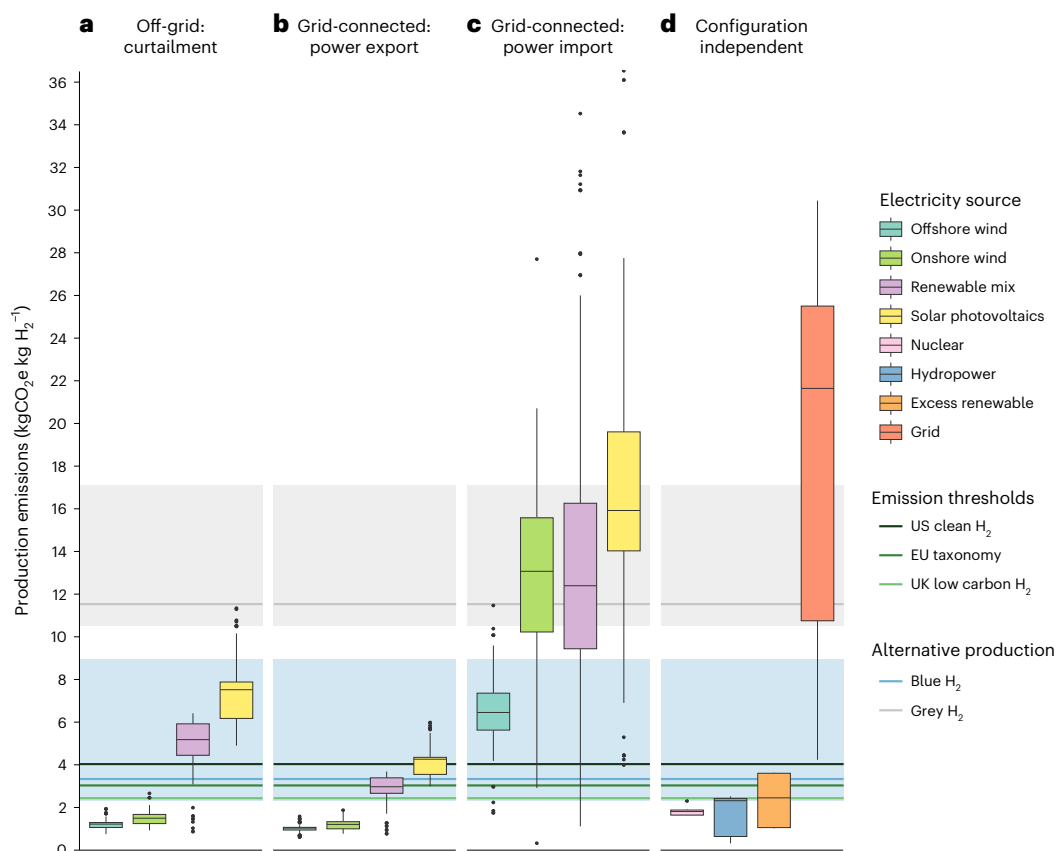


Fig. 2 | Relation between green hydrogen production GHG emissions and electricity source and production configuration. a–d. The shaded areas indicate emission ranges for hydrogen production from steam methane reforming (grey H_2) and from steam methane reforming combined with 93% CO_2 capture and storage (blue H_2). The range represents upstream methane leakage rates of 0.2% (lower bound) to 8% (upper bound), with a horizontal line at 1.5% (representative mid-range value)³⁴. The horizontal green lines show cradle-to-gate emissions thresholds for hydrogen production¹⁰. Interquartile range and

median of cradle-to-production gate emissions of hydrogen ($kgCO_2e\ kg\ H_2^{-1}$) are shown by intermittent renewable electricity source in the ‘off-grid: curtailment’ configuration (a), ‘grid-connected: power export’ configuration (b) and ‘grid-connected: power import’ configuration (c) and by configuration-independent electricity source (d). The whiskers represent all values within 1.5 times the interquartile range, and outliers are shown as black dots. See Supplementary Table 11 for underlying values.

Transport via pipelines leads to the lowest emissions across transport options on short distances, adding an extra $1.1\ kgCO_2e\ kg\ H_2^{-1}$ for a distance of 500 km, and $1.5\ kgCO_2e\ kg\ H_2^{-1}$ for a distance of 1,000 km, as the median across projects. Conversely, for 5,000 km transport, pipelines lead to the highest median emissions across transport options, adding $5.2\ kgCO_2e\ kg\ H_2^{-1}$ (Fig. 4b). The steep increase in emissions results from the recompression of gaseous hydrogen approximately every 125 km (ref. 15) to maintain the pressure along the pipeline, and the estimated 0.5% hydrogen loss per (re-)compression^{33,35}. While electricity for initial compression is covered by the same source as for electrolysis, the assumed GHG intensity for electricity in booster stations is $37\ gCO_2e\ kWh^{-1}$, the calculated global average for wind and solar electricity. Hydrogen loss during recompression means that (1) more hydrogen must be produced for 1 kg of hydrogen to arrive at the destination, increasing production emissions by $0.1\text{--}0.7\ kgCO_2e\ kg\ H_2^{-1}$ for the 500–5,000 km range (Fig. 4b and Supplementary Fig. 4), and (2) more hydrogen leaks into the atmosphere, which, based on an estimated 100-year global warming potential of $11\ kgCO_2e\ kg\ H_2^{-1}$ (refs. 36,37), adds $0.3\text{--}2.3\ kgCO_2e\ kg\ H_2^{-1}$ delivered over a 500–5,000 km distance. Pipeline transport emissions therefore depend on the hydrogen project characteristics in two ways: (1) the electricity source for initial compression, and (2) increased production to make up for losses. Pipeline transport emissions are relatively sensitive to assumptions made. If compressor losses can be reduced from 0.5% to 0.2% (ref. 35), for instance, and the

interval between recompression stations can be extended to 500 km (ref. 38), pipeline transport emissions could be reduced by 27–58% to $0.8\text{--}2.2\ kgCO_2e\ kg\ H_2^{-1}$ over a 500–5,000 km distance.

Shipping liquid hydrogen would cause less emissions than shipping hydrogen that has been converted to ammonia and is reconverted at the destination (‘cracking’) (Fig. 4b), in line with earlier findings¹⁸. Emissions associated with the transport supply chain are, however, substantial in both cases, with $1.7\text{--}4.9\ kgCO_2e\ kg\ H_2^{-1}$ over a 500–20,000 km distance for liquid hydrogen and $4.2\text{--}6.1\ kgCO_2e\ kg\ H_2^{-1}$ over a 500–20,000 km distance for ammonia with cracking (medians across projects). Hydrogen liquefaction itself is an electricity-intensive step and contributes $0.7\ kgCO_2e\ kg\ H_2^{-1}$ (median across projects), using the same electricity source as the electrolyser. A possible future reduction of the electricity requirement from 12 to $9\ kWh\ kg\ H_2^{-1}$ (refs. 39,40) still results in median liquefaction emissions of $0.6\ kgCO_2e\ kg\ H_2^{-1}$. Hydrogen boil-off losses resulting from cryogenic conditions during shipping¹⁶ cause an increase in overall transport emissions, both from leaked hydrogen’s direct role as a GHG ($0.4\text{--}1.3\ kgCO_2e\ kg\ H_2^{-1}$ delivered over 500–20,000 km) and ultimately increased production emissions to cover losses ($0.1\text{--}0.4\ kgCO_2e\ kg\ H_2^{-1}$ for the 500–20,000 km range). For ammonia transport, the conversion of hydrogen to ammonia emits $1.8\ kgCO_2e\ kg\ H_2^{-1}$ delivered, and cracking to recover hydrogen emits $0.3\ kgCO_2e\ kg\ H_2^{-1}$. In addition, to make up for losses in conversion and cracking (which is assumed to recover 69.5% of the hydrogen content¹⁸), hydrogen production needs to increase by

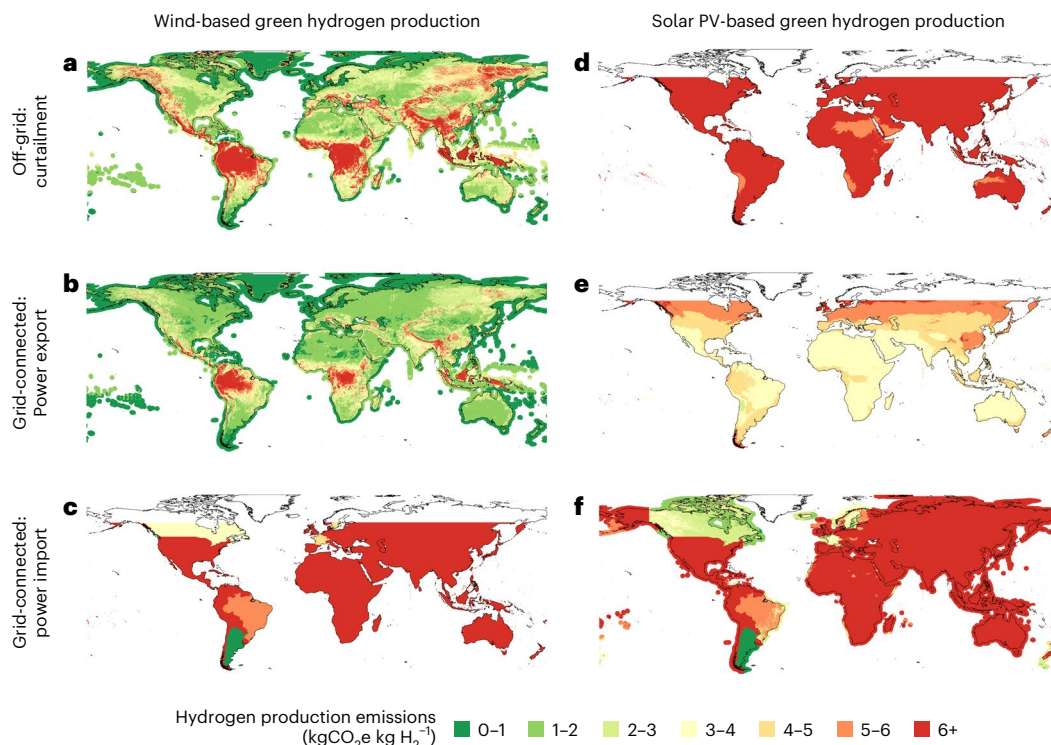


Fig. 3 | Spatial variation in cradle-to-production gate GHG emissions of green hydrogen produced with wind and solar power. a–c, Emissions are shown for hydrogen produced in a PEM electrolyser, using onshore and offshore (up to 200 nautical miles off the coast) wind power in the ‘off-grid: curtailment’ (a), ‘grid-connected: power export’ (b) and ‘grid-connected: power import’ (c) configurations. **d–f,** Solar photovoltaic (PV)-based hydrogen production emissions are shown for the ‘off-grid: curtailment’ (d), ‘grid-connected: power export’ (e) and ‘grid-connected: power import’ (f) configurations. Emissions are based on regression model-derived GHG footprints for renewables^{23,24} using local meteorological conditions^{65,67}.

(c) configurations. **d–f,** Solar photovoltaic (PV)-based hydrogen production emissions are shown for the ‘off-grid: curtailment’ (d), ‘grid-connected: power export’ (e) and ‘grid-connected: power import’ (f) configurations. Emissions are based on regression model-derived GHG footprints for renewables^{23,24} using local meteorological conditions^{65,67}.

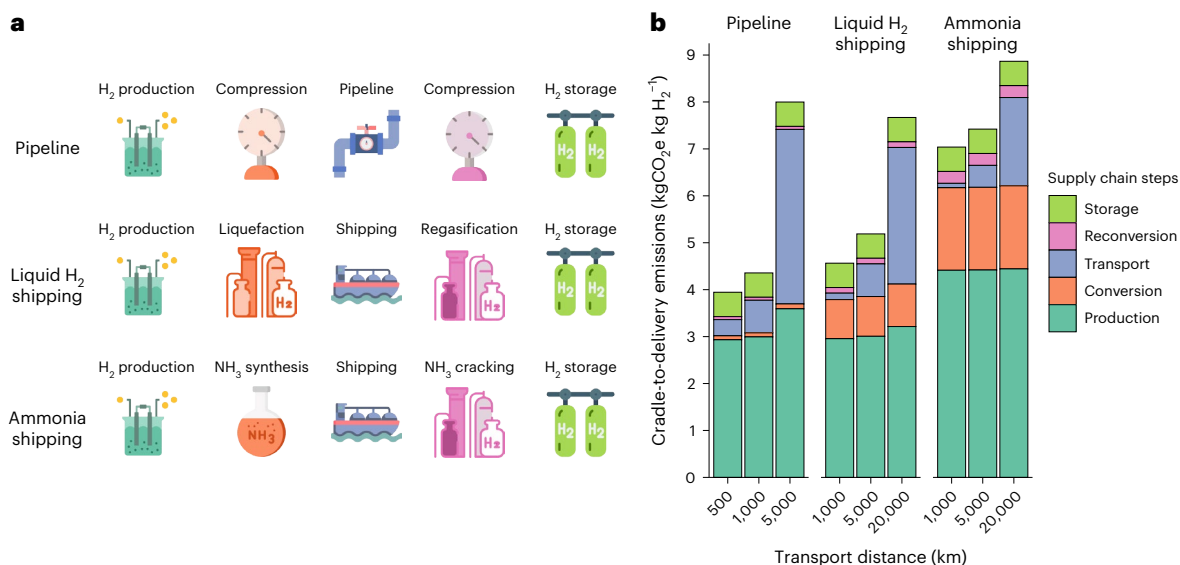


Fig. 4 | Contributions of production, conversion, transport, reversion and storage to cradle-to-point of delivery emissions of green hydrogen. a, Steps in the supply chain for production and transport of hydrogen by pipeline or ship as liquid hydrogen or ammonia. **b,** Median values ($\text{kgCO}_2\text{e kg H}_2^{-1}$ delivered) across all projects (excluding grid electricity based) in the ‘power

export’ configuration for transport distances of 500 km, 1,000 km and 5,000 km for pipelines and 1,000 km, 5,000 km and 20,000 km for shipping. See Supplementary Fig. 5 for emissions of ammonia without cracking at the destination.

55% to deliver the same amount of H_2 compared with the situation of hydrogen production without conversion to ammonia and back. This adds $1.6 \text{ kgCO}_2\text{e kg H}_2^{-1}$, resulting in total emissions at the production stage of $4.4 \text{ kgCO}_2\text{e kg H}_2^{-1}$ delivered (Fig. 4b and Supplementary Fig. 4).

Emissions from liquid hydrogen shipping itself ($0.07\text{--}2.9 \text{ kgCO}_2\text{e kg H}_2^{-1}$ over a 500–20,000 km distance) are about twice as high as for ammonia ($0.03\text{--}1.3 \text{ kgCO}_2\text{e kg H}_2^{-1}$ over a 500–20,000 km distance), because more ships are needed due to liquid hydrogen’s low volumetric density.

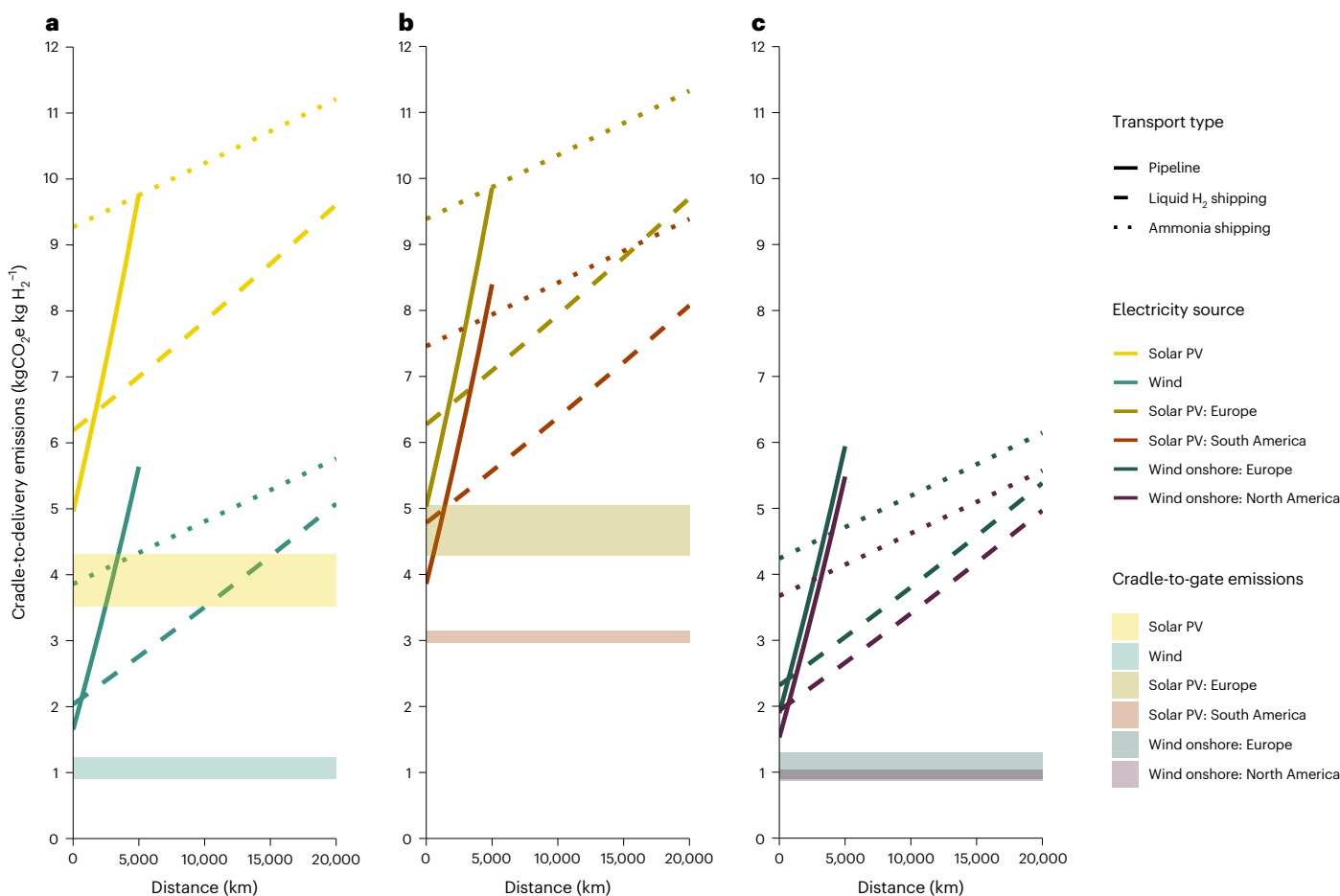


Fig. 5 | GHG emissions of green hydrogen production and transport over increasing transport distances. a–c, Median cradle-to-point of delivery emissions compared with the interquartile range of cradle-to-production gate emissions indicated by shaded areas ('power export' configuration): across all projects using wind electricity (onshore and offshore combined) or solar PV electricity (a); across solar-based projects in Europe and South America only (b); and across wind-based projects in Europe and North America only (c). Note

that cradle-to-point of delivery emissions for a transport distance of 0 km are higher than cradle-to-production gate emissions due to contributions after the production gate that do not scale with distance. These processes include initial compression, liquefaction and conversion to ammonia, ammonia cracking and storage of hydrogen. The corresponding emissions are included in the cradle-to-point of delivery emissions at a distance of 0 km.

For all transport options, storage infrastructure is probably required, adding 0.5 kg CO₂e kg H₂⁻¹ for 3 days' worth of storage capacity.

Including the transport of green hydrogen in the footprint calculation is critical for understanding the electricity-source-dependent trade-offs between transport options and production locations. Generally, wind-based production has much lower emissions than solar-based production. Beyond a specific (transport option-dependent) distance, however, emissions from transported wind-based hydrogen exceed those of locally produced solar-based hydrogen (Fig. 5a). Another trade-off exists between producing solar-based hydrogen close to demand at higher emissions in places with less solar irradiation, versus lower-emission production in locations with higher solar irradiation and then transporting it. Transport from the best solar locations could only compete with local production if the transport distance does not exceed 2,000 km, due to the emissions associated with transport (Fig. 5b). Producing in low-irradiation locations in Europe, for instance, still yields lower emissions than importing from high-irradiation locations in South America. Long-distance transport of wind-based hydrogen always leads to higher emissions than local wind-based production in our analysis, even if hydrogen is produced in the best wind locations and then transported to places with the lowest wind potential (Fig. 5c), as the difference in production emissions is too small to make up for transport emissions.

Mitigation potential of planned green hydrogen projects

Using our GHG emission calculations and the 2030 hydrogen production rates of each project²², we derive a global emission–supply curve for green hydrogen production (Fig. 6a). In the 'power export' configuration, 55% of projects and 85% of cumulative production are below the EU taxonomy's threshold (3.0 kg CO₂e kg H₂⁻¹), and 89% of projects and 95% of cumulative production are below the US threshold (4.0 kg CO₂e kg H₂⁻¹). In the 'curtailment' and 'power import' configurations, 39% and 21% of cumulative production, and 25% and 9% of all projects, respectively, are below the EU threshold; 57% and 23% of cumulative production, and 36% and 17% of all projects, respectively, are below the US threshold. All planned hydrogen projects together would produce 65 million tonnes of hydrogen in 2030, emitting 0.16 giga tonnes of CO₂e for production only ('power export' configuration). Compared with producing the same amount of grey hydrogen, planned green hydrogen production would avoid 0.6 Gt CO₂e yr⁻¹ (0.5 and 0.1 Gt CO₂e in the 'curtailment' and 'power import' configurations, respectively). This replacement of existing applications of fossil-fuel-based hydrogen is considered a priority⁴¹ and fits within the current grey hydrogen market of 100 Mt H₂ per year⁸, but other uses of green hydrogen may result in different levels of avoided emissions.

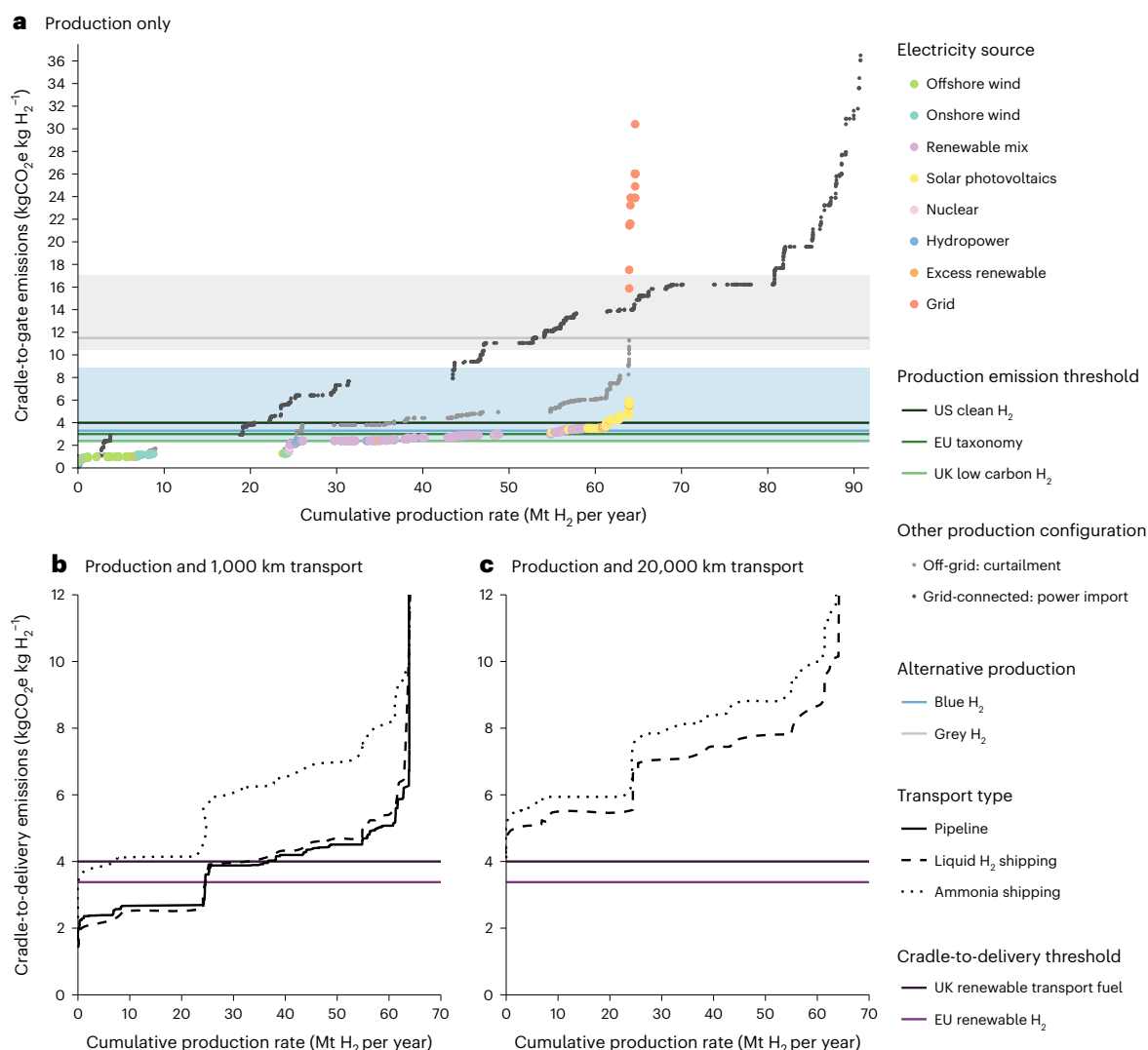


Fig. 6 | GHG emission-supply curves for green hydrogen. Emission-supply curves show the cumulative potential supply of hydrogen at increasing levels of GHG emission intensity. The horizontal lines and shaded areas indicate emissions of alternative production technologies and policy-based emission thresholds^{10,34,84}. **a**, Cradle-to-production gate emission-supply curve in colour

for the 'power export' configuration, and grey and black for the 'curtailment' and 'power import' configurations, respectively. **b,c**, Emission-supply curves for different transport modes in the 'power export' configuration, for hydrogen production and 1,000 km of transport (**b**) and for production and 20,000 km of transport (**c**).

When including 1,000 km of transport in the emissions supply curves, total emissions become higher, at 0.26 GtCO_2e for pipeline or liquid hydrogen, or 0.40 GtCO_2e for ammonia transport. About 37 Mt of hydrogen per year could be produced and transported over a distance of 1,000 km at emissions below the UK renewable fuel threshold ($4.0 \text{ kg CO}_2\text{e kg H}_2^{-1}$), 25 Mt below the EU threshold ($3.38 \text{ kg CO}_2\text{e kg H}_2^{-1}$; Fig. 6b; note that these thresholds include hydrogen distribution). There are virtually no projects for which transport as ammonia would stay below any of the emissions thresholds. Furthermore, no projects would stay below emission thresholds for transport distances over 20,000 km (Fig. 6c).

The European Commission's policy target for 2030 is 10 Mt of green hydrogen production in Europe and 10 Mt of imports. Based on the planned projects for domestic production, wind-based hydrogen (6.2 Mt in Europe) would stay below the EU threshold in the 'power export' and 'curtailment' configurations, whereas solar-based production (2.1 Mt) would not. For 10.9 Mt of planned European production, the renewable electricity source is still unknown (Supplementary Fig. 6). Based on planned projects outside of Europe, the import target

of 10 Mt of green hydrogen cannot be fulfilled while staying below the emission threshold. Options below the threshold include wind-based production in North Africa with transport to Europe via pipelines or as liquid hydrogen, or hydropower-based production in North America with transport as liquid hydrogen, but the announced projects of this type do not add up to 10 Mt.

If 300 to 800 Mt of green hydrogen were to be produced annually in 2050 (as some scenarios project for 2050 (ref. 8)), and if we assume the same distribution of production emissions as in the 'power export' configuration (but excluding grid-based projects), annual emissions of green hydrogen production would range from 0.7 to 1.8 GtCO_2e , excluding transport. Production at this scale would require 15–39 PWh of renewable electricity (assuming no curtailment), requiring an unprecedented growth compared with 1.3 PWh of solar photovoltaic electricity and 2.1 PWh of wind electricity generated worldwide in 2022 (ref. 42). Taking a scenario of 1/3 of all green hydrogen being used on site, 1/3 being transported 1,000 km by pipeline and 1/3 being shipped as liquid hydrogen over 10,000 km, this would add a further 0.4–1.2 GtCO_2e of annual GHG emissions. Overall, this is equivalent to

about 1.2–3.3% of current global GHG emissions⁴³, or about 2.1–5.5% when including the transport scenario. Therefore, achieving net-zero targets for 2050 (ref. 44) requires not only the scale-up of green hydrogen production but also a substantial reduction of emissions along the green hydrogen supply chain in parallel.

Discussion and conclusions

Securing investment for export-oriented green hydrogen projects requires clarity on sustainability standards¹⁹, and it is critical to get these standards right for the hydrogen economy to lead to the envisioned emission reductions. Current regulations have a large blind spot: counting renewable electricity as zero emissions (for example, EU Delegated Act on Article 27 of the RED II⁴⁵) and excluding emissions from component manufacturing and leakage (for example, US Clean H₂ Standard⁴⁶). Both make it deceptively simple to stay below emissions thresholds for green hydrogen production, as opposed to when full life-cycle GHG emissions are included.

Including life-cycle GHG emissions shows that it is imperative that the electricity used for hydrogen production has a low GHG intensity over the whole supply chain. As long as electricity grids are not fully decarbonized yet, using electricity from the grid, even if only as a supplement to intermittent renewables, can result in emissions exceeding those of grey hydrogen production, undermining the climate mitigation potential of such projects. For grid-connected green hydrogen projects, it is therefore crucial that the additionality of renewable energy facilities can be guaranteed. Without strictly enforced additionality, electrolyzers would effectively run on the average grid mix, with associated emissions⁴⁷. 'Simply' claiming to use intermittent renewable electricity obscures challenges around balancing operational hours of the electrolyser, the relative size of the renewable power generation facility and choices regarding the inclusion of batteries, a grid connection and electricity curtailment. We show that successfully embedding the electrolyser in the power system and securing buyers of surplus electricity may be critical to producing green hydrogen at the lowest possible life-cycle emissions.

It is over-simplistic to assume zero emissions for renewable electricity and, therefore, important to distinguish between types of renewable: whereas hydrogen production using solar electricity would exceed emission thresholds in most locations, production based on hydropower or wind power would not. As long as the manufacturing process of solar panels and wind turbines, electrolyzers and batteries (partly) relies on fossil energy, the production of hydrogen will always induce emissions. Furthermore, life-cycle emissions of hydrogen are greatly increased by transport—depending on transport mode and distance and on whether or not losses along the way can be minimized. Production closer to demand generally outperforms the benefit of producing at lower emissions in far-away locations with favourable weather conditions. Transport emissions are thus key to understanding the benefits and trade-offs of different locations and should be considered in the system boundaries of standards. Hydrogen's global warming potential plays a non-negligible role where transport losses occur and should therefore also be included. In this analysis, no hydrogen leakages in the production step are included, although they may be included in future analyses when more data become available.

Although average production emissions across projects are below the EU's threshold for sustainable hydrogen production, total life-cycle emissions of projected hydrogen production would still be substantial and, hence, unaligned with absolute sustainability⁴⁸. There is a trade-off between scaling up the green hydrogen economy, which in the short term may be hindered by too strict criteria for initial projects, and the need for sufficiently strict and comprehensive emission thresholds for achieving necessary emission reductions in the medium and long term⁴⁹. Simultaneously, attention is needed for risks for energy injustices associated with the roll-out of the hydrogen economy⁵⁰,

for example, hydrogen production for export at the expense of local use of renewable capacity⁵¹.

Our results demonstrate that efforts are needed to lower green hydrogen production emissions, including from the manufacturing of electrolyzers, storage infrastructure and renewable energy technologies—which may, in fact, depend on low-emission hydrogen availability for hydrogen-based steelmaking⁵². Investigating the use of salt caverns for hydrogen storage could present an economically and environmentally beneficial alternative to hydrogen storage tanks, albeit contingent on the proximity of these caverns to hydrogen production sites⁵³. While large parts of the upstream supply chain are probably beyond the feasible scope of hydrogen production standards, it is nevertheless important that the implied emissions and trade-offs are considered in policy discussions on how to facilitate the hydrogen market ramp-up, for example, in the context of hydrogen import strategies. Although emissions from some production processes (such as steelmaking) are already covered by other policies (such as the EU Emissions Trading Scheme), this only applies in certain jurisdictions. This could justify their parallel consideration in hydrogen standards, in particular when it comes to global hydrogen trade. Furthermore, increasing electrolyser efficiency and operational hours, increasing hydrogen conversion and reconversion efficiencies, avoiding losses during transport and storage, or minimizing transport altogether by co-locating industry with hydrogen production^{54,55} are additional avenues to explore.

Methods

Software used for the analysis

All analyses were performed in R (version 4.1.2), R Studio (version 2023.06.0), ArcGIS Pro (version 3.1.2) and Simapro (version 9.4.0.2), using EcolInvent (version 3.8) and ReCiPe 2016 Midpoint (H) V1.07.

Selection of projects from the IEA database for the analysis

Out of the 1,477 hydrogen production projects in the IEA Hydrogen Projects database (version of October 2022)²², 1,025 were included in our analysis, based on the following selection criteria. (1) Hydrogen production must be based on electrolysis. (2) The project is expected to be operational in 2030. Therefore, projects decommissioned before 2030 or starting after 2030 are excluded. If no starting data are indicated, we assumed the starting date is 2030. (3) The size of the project is indicated. (4) The location is indicated. The database includes information on project-specific electricity source to power electrolysis, electrolyser technology, country and project size (Supplementary Fig. 1).

Electrolyser energy and material requirements

The IEA database²² distinguishes between alkaline (ALK), polymer electrolyte membrane or proton exchange membrane (PEM), solid oxide electrolysis cell (SOEC) electrolyzers and 'other electrolysis', each with different energy and material requirements. We use the life-cycle inventory for ALK, PEM and SOEC electrolyzers by Gerloff¹³ as the basis of our analysis, including construction of the electrolyser stacks and 'balance of plant'¹³ (Supplementary Table 2). We adapted the electricity demand by using the expected 2030 values instead of the 2017 values from the same study⁵⁶. The SOEC has an output pressure of 1 bar (ref. 17), so its electricity requirement includes an additional compression step to reach the same output pressure as the ALK/PEM electrolyzers (30 bar)^{14,28}. For 'other electrolysis', we calculate the average of the requirements of the ALK, PEM and SOEC electrolyzers.

We assume that SOEC electrolyser heat demand is met using natural gas⁵⁷ (EcolInvent process 'market group for heat, district or industrial, natural gas {GLO}'), as this has lower emissions than the EcolInvent process 'market group for heat, district or industrial, other than natural gas {GLO}'¹³. The GHG intensity of the electricity requirement is calculated on the basis of the electricity source and location documented in the IEA database²². We use country-specific information, while for

six projects where multiple countries are given as the location, their average is used.

As many of the planned projects are located in water-stressed regions⁸, we assume that sea water desalination is required to supply the water used for electrolysis (feed and cooling water) proportional to the local level of water stress²⁷. The electricity requirement for desalination is around 3.8 kWh per cubic metre of water^{28,58}. Landlocked countries (not bordering a sea) are excluded from the sea water desalination requirement. As a result, the electricity and non-electricity emissions for water desalination are multiplied by a factor between zero and one. The factor is zero for landlocked countries and where there is no water stress, and the factor is one where water stress is highest. After desalination, water is deionized, for which we adapt the relevant EcoInvent processes based on Terlouw et al.¹⁴. Water deionization is included for all projects, regardless of water stress level. Required cooling water¹³ is assumed to be recirculated five times to reduce water demand and thermal discharge into surface water⁵⁹.

Green hydrogen production configurations

We assume continuous operation of the electrolyser for grid-powered systems¹⁴, as well as for hydropower and nuclear electricity sources. The utilization factor of the electrolyser ($f_{\text{utilization}}$) is equal to 1 for continuous operation. We use the documented²² electrolyser capacity or annual hydrogen production rate to calculate the missing parameters using equations (1) and (2):

$$M_{\text{H}_2} = \frac{\text{cap}_{\text{electrolyser}} \times f_{\text{utilization}} \times 8,760 \text{ h}}{P_{\text{electrolyser}}} \quad (1)$$

$$\text{cap}_{\text{electrolyser}} = \frac{M_{\text{H}_2} \times P_{\text{electrolyser}}}{f_{\text{utilization}} \times 8,760 \text{ h}} \quad (2)$$

where M_{H_2} is the annual hydrogen production rate in tonnes of H_2 per year, $\text{cap}_{\text{electrolyser}}$ the installed electrolyser capacity in MW and $P_{\text{electrolyser}}$ the electricity requirement for electrolysis in kWh per kilogram of H_2 (Supplementary Table 2).

Hydrogen production systems operated by solar or wind electricity have an $f_{\text{utilization}}$ lower than 1 due to the intermittent character of these sources. Based on energy system optimization, cost-optimal configurations for dedicated renewable systems have an $f_{\text{utilization}}$ between 0.5 and 0.6 (ref. 14), which means that, over the course of the year, 0.5–0.6 times the hydrogen is produced of continuous production. In these configurations, electricity generation facilities are oversized in combination with battery storage and curtailment of electricity. The energy-to-power ratio of the battery is between 2 and 10 (mean 6), which expresses how many hours' worth of electricity can be stored for full electrolyser operation. Despite the use of a battery that allows for production of hydrogen with a time lag from when the electricity is produced, curtailment of electricity is high, with a curtailment factor between 0.2 and 0.7 (mean 0.5). An alternative configuration is characterized by an $f_{\text{utilization}}$ of 0.4, a battery that can store 3 hours' worth of electricity for full electrolyser operation and no curtailment²⁸.

When electricity must be curtailed, emissions from manufacture of the full scale of the electricity facilities are allocated to hydrogen production. In this case, there is no grid connection or no user of this electricity ('off-grid: curtailment' configuration). If a grid connection is in place and the surplus is used by other economic activities, these emissions need not be allocated to hydrogen ('grid-connected: power export' configuration) (Table 1). Furthermore, we include a third configuration representing a grid-connected system with a utilization factor of 1, where the renewable generation capacity equals the electrolyser capacity¹⁴ and powers the electrolyser for the share of the year equal to the country-specific capacity factor (75th percentile). The difference between the renewable capacity factor would be supplied by the grid in this 'grid-connected: power import' configuration (Table 1).

Since the specific optimal configuration of energy generation oversizing, battery size and curtailment depends on hourly¹⁴ or even minute-to-minute²⁸ electricity generation profiles and local costs of the respective components, it is out of the scope of this paper to optimize this for each specific project. Instead, we use hydrogen production configuration parameters based on these earlier studies^{14,28} (Supplementary Table 3) and assess the choice of parameters in the sensitivity analysis.

When electrolysis is powered by excess renewable electricity, we assume that the electrolyser operates only when excess electricity is available. Only when intermittent renewables make up more than 50–60% of the electricity mix is the curtailment factor higher than 0.025 (ref. 31). For Germany, with relatively high renewables penetration, 471 surplus hours are foreseen in 2032 (ref. 30), which means that $f_{\text{utilization}}$ would be 0.05, which we use for all excess-based projects.

Battery manufacturing emissions

The battery is assumed to be a lithium-ion battery with a roundtrip efficiency of 90% (refs. 14,28). Since the theoretical maximum of the share of electricity that passes through the battery before arriving at the electrolyser is 0.5, the gross electricity generation requirement is at most 5% higher than the net electricity requirement. Given that weather patterns are much more variable than required for this maximum, we assume that 0.25 of electricity passes through the battery. We use the GHG emissions of lithium nickel manganese cobalt oxide battery manufacture (including the battery system with pack, battery management system, energy management system, battery thermal management system, system container, inverter and transport⁶⁰), estimated at 68.6 kgCO₂e kWh⁻¹ of storage capacity in 2030 in a 2 °C scenario⁶¹.

Emissions of grid electricity

To take the decarbonization of grid mixes in the future into account, we use the GHG intensities of modelled 2030 grid mixes per country for a 2 °C policy scenario²⁶. When a country is not modelled separately but as part of a region, we use the GHG intensity of the region it belongs to.

Emissions of hydroelectricity

The GHG intensity of hydropower depends on the size of the facility and facility type, that is, impoundment (reservoir and dam) or diversion (run of river)²⁵. Although the IEA database²² does not include information on the facility type for projects using hydropower, we used the project name and sources in the database to find the facility type and match it with the mean GHG intensity: 21.1 gCO₂e kWh⁻¹ for small impoundment systems (0.1–30 MW); 40.6 gCO₂e kWh⁻¹ for large impoundment systems (>30 MW); 27.2 gCO₂e kWh⁻¹ for small diversion systems (0.1–30 MW); and 3.5 gCO₂e kWh⁻¹ for large diversion systems (>30 MW)²⁵. The higher emissions for large impoundment systems are a result of construction requirements and flooding of land.

Emissions of nuclear electricity

Relatively large GHG ranges have been reported for nuclear electricity: 10–130 (ref. 62) or 3.7–110 gCO₂e kWh⁻¹ (ref. 63), explained by different GHG intensities of the background economy, different uranium ore grades and different enrichment methods. We use the more recent value of 27.6 gCO₂e kWh⁻¹ found using a hybrid life cycle assessment approach⁶⁴.

Emissions of offshore and onshore wind electricity

We created a generalized linear model (equation (3)) based on wind turbine data collected and analysed by Dammeier et al.²⁴ to estimate the GHG intensity of wind electricity based on solely wind speed and onshore or offshore location:

$$\log_{10}(\text{EF}_{\text{GHG}}) = 2.2571 - 0.025 \times v_{\text{avg}} + 0.3066 \times O \quad (3)$$

where EF_{GHG} is the emissions intensity of wind electricity ($\text{gCO}_2\text{e kWh}^{-1}$), v_{avg} is the 30-year average wind speed at 100 m in m s^{-1} and O is a categorical variable denoting onshore (0) or offshore (1) wind farms. The model's R^2 is 0.84, and end-of-life emissions of the wind farms are not included.

Wind speed is location dependent. The GHG intensity of wind electricity is estimated on the basis of onshore or offshore location and wind speed, using data from the GlobalWindAtlas⁶⁵, based on ERA5 data for the years 2008–2017. We use the exclusive economic zone to determine the offshore wind speed, which is the area up to 200 nautical sea miles from the coast. We use the 75th percentile of wind speed at the country level or in the exclusive economic zone, assuming that the facility is built in a favourable location. In the sensitivity analysis, we use the median and 90th percentile of wind speed.

Emissions of solar photovoltaic electricity

Based on a regression model requiring only panel type, yearly irradiation and age of a facility by Bosmans et al.^{23,66}, the GHG intensity of solar electricity can be estimated using equation (4):

$$\begin{aligned} \log(EF_{\text{GHG}}) &= 64.3 - 0.03088 \times Y \\ &= \text{year} + \beta_{\text{type}} - 0.000232 \times I_{\text{mean}} + \beta_{Y\text{-type}} \times \text{year} \end{aligned} \quad (4)$$

where EF_{GHG} is the emissions intensity of solar electricity ($\text{gCO}_2\text{e kWh}^{-1}$), year is the absolute value and I_{mean} is irradiation in $\text{kWh m}^{-2} \text{yr}^{-1}$. β_{type} is -12.7 for CdTe, -49.4 for Cl(G)S, 48.9 for mono-Si, 47.6 for poly-Si and 0 for a-Si; $\beta_{Y\text{-type}}$ is 0.0062 for CdTe, 0.025 for Cl(G)S, 0.024 for mono-Si, 0.024 for poly-Si and 0 for a-Si. This model has an R^2 of 0.99, and end-of-life emissions of the solar photovoltaic facilities are not included. Further details on estimated panel type modelling is included in Supplementary Information section 3.

Irradiation is location dependent. The GHG intensity of solar electricity is estimated on the basis of global horizontal irradiation at the country level, using data from the GlobalSolarAtlas⁶⁷. The GlobalSolarAtlas is based on a SolarGIS model using several long-term datasets, together providing a climatic mean⁶⁸, and covers latitudes from parallel 45°S to parallel 60°N . No hydrogen projects based on solar photovoltaic in the IEA database²² are located outside this range. The country averages for countries passing through these parallels are based on the data that fall into the range (for example, Sweden, Canada, Argentina and Chile). We take the 75th percentile of the area's irradiation, assuming that the facility is built in a climatically favourable location and can feasibly be connected to the grid⁶⁹. In the sensitivity analysis, we use the median and 90th percentile of solar irradiation.

Emissions of excess renewable electricity

Hydrogen production has been proposed as a power-to-X technology, using electricity when the supply of electricity temporarily exceeds demand. Some studies include the GHG intensity of renewable electricity generation for such electricity^{58,70–73}, while other studies assume a zero GHG intensity for excess electricity arguing that the excess would have a zero cost or would otherwise have been wasted^{31–33,74–77}. We assume a GHG intensity of $0 \text{ kgCO}_2\text{e kWh}^{-1}$ in the default calculations and the country-specific mean of solar and wind electricity in the sensitivity analysis (a shared contribution of solar photovoltaic (50%) and wind (50%)—half of it offshore and half onshore).

Emissions for unknown electricity sources

When projects lack information on the electricity source used, we assume a shared contribution of solar photovoltaic (50%) and wind (50%)—half of it offshore and half onshore.

Green hydrogen transport via pipeline

Our analysis of hydrogen transport via pipeline is based on a life cycle assessment study that includes thermodynamic modelling of hydrogen

transport via new steel pipelines with a diameter of 48 inches and operating at 80 bar (ref. 15). We assume the pipeline is built at or near the H_2 production location. As the pipeline network can be a storage solution¹⁵, hydrogen storage before transport is not needed. Since hydrogen is produced at lower pressure than required for the pipes, hydrogen is first compressed, using the same electricity source as used for electrolysis. Furthermore, the pressure drops along the pipeline and recompression is required every 125 km to reduce environmental impacts¹⁵ and costs⁷⁸. The distance between recompression stations depends on the hydrogen mass flow rate and pipeline diameter, where smaller flow rates and larger pipeline diameters correspond to larger distances between stations. We use the reference case of 125 km (ref. 15) as the default, and test the maximum distance of 500 km (ref. 38). We include leakage of hydrogen at the compressor station, which is estimated to be 0.5% (ref. 33). New technologies may limit this loss to 0.2% (ref. 35), which we assess in an additional scenario. We assume that a final recompression step is needed before storage of gaseous hydrogen at the destination at 80 bar, and include the infrastructure requirement for 3 days' storage capacity^{14,28}. Electricity requirements for recompression are covered by a global average of solar and wind electricity, with a GHG intensity of $37 \text{ gCO}_2\text{e kWh}^{-1}$. Further details on pipeline transport are provided in Supplementary Table 6.

Hydrogen liquefaction and liquid hydrogen shipping

We assume that the production plant is situated close to a port and includes a storage capacity on land equal to the capacity of the tanker or to 21 days' worth of production capacity⁷⁹ and that the electricity for post-electrolysis processing is met by the same electricity source as electrolysis. Gaseous hydrogen is liquified to increase the density for shipping. Industrial hydrogen liquefaction currently consumes between 12.5 and 15 kWh kg H_2^{-1} (refs. 39,40). For 2030, we use 12 kWh and include a possible decrease to 9 kWh (refs. 39,40) in an additional scenario. We use a LCI of a 50 tH_2 per day liquefaction plant^{29,79} and scale it to emissions per kilogram of H_2 for each project. After liquefaction, liquid H_2 is stored at -253°C (refs. 40,79), loaded, shipped, unloaded, regasified and compressed, and stored as gaseous hydrogen for 3 days. We further include evaporation of liquid hydrogen (boil-off gas; see 'Boil-off gas generation for liquid energy carriers' section). For electricity used at the destination, we assume a global average of solar and wind electricity, with a GHG intensity of $37 \text{ gCO}_2\text{e kWh}^{-1}$. We adapted the EcoInvent entry for liquified natural gas sea transport to liquid H_2 shipping²⁹ (see Supplementary Tables 7 and 8 for further details on liquid hydrogen transport).

Hydrogen conversion to ammonia and ammonia shipping

We assume that the production plant is situated close to a port and includes a storage capacity on land equal to the capacity of the tanker or to 21 days' worth of production capacity⁷⁹, and that the electricity for post-electrolysis processing is met by the same electricity source as electrolysis. For (green) ammonia production, we consider a flexible, electricity-based Haber–Bosch ammonia production process^{80,81}, using N_2 from a cryogenic air separation unit for which we adapt the EcoInvent process to harmonize the electricity source to be equal to that of electrolysis. In case of variable hydrogen production, that is, based on intermittent renewable electricity, the ammonia production process requires 2 days of hydrogen storage to buffer fluctuations⁸¹, based on the inventory for gaseous hydrogen storage vessels^{14,28}. Ammonia is liquified and temporarily stored, for which we adapt the inventory for hydrogen storage vessels to the much higher volumetric density of liquid NH_3 . Boil-off gas during storage is reliquefied and recirculated¹⁸. We adapt the EcoInvent entry for liquified natural gas sea transport for ammonia shipping²⁹. At the receiving port, ammonia is unloaded and temporarily stored (1 day). For electricity used at the destination, we assume a global average of solar and wind electricity, with a

GHG intensity of 37 gCO₂e kWh⁻¹. Ammonia cracking is an endothermic reaction requiring temperatures of 500 °C or higher. The process uses electricity and part of the hydrogen as fuel, which is assumed to recover 69.5% of the hydrogen content¹⁸. Then, 3 days of gaseous hydrogen storage are additionally included. In an additional scenario presented in Supplementary Information section 10, ammonia is delivered to the market as ammonia, without cracking. In that case, 3 days of storage capacity are included. See Supplementary Tables 7 and 9 for further details on the ammonia transport.

Boil-off gas generation for liquid energy carriers

When a liquid energy carrier is stored under cryogenic conditions, that is, at a temperature below its boiling point, boil-off gas (BOG) is produced due to the difference in temperature between the liquid and the environment¹⁶. While during storage the gas is assumed to be reliquified and recirculated to storage^{18,29}, during loading, unloading and shipping the BOG is generally considered to be evaporated to the atmosphere and counted towards an energy or mass loss^{29,39,40,82}. BOG during shipping depends on the duration of shipping and is calculated on the basis of the shipping distance, assuming a speed of 16 knots (Supplementary Tables 8 and 9). Although in the future it might be possible to use BOG as part of the shipping fuel, this process still needs to be developed^{16,18,39,82} and heavy fuel oil is typically assumed to be used as transport fuel for shipping liquid hydrogen and ammonia in 2030 (refs. 29,40,80), which is what we assume in our analysis.

Construction emissions and lifetime

The construction emissions of all technology components are scaled to the installed electrolyser capacity or annual hydrogen production rate for each project. Construction emissions are divided over the total amount of hydrogen produced in the project following

$$EF_{\text{construction}} = \frac{EM_{\text{construction}}}{\text{Lifetime} \times M_{H_2}} \quad (5)$$

where $EF_{\text{construction}}$ is the emissions from construction in kgCO₂e kg H₂⁻¹ produced, $EM_{\text{construction}}$ is the total emissions from construction of the component in kgCO₂e, Lifetime is equipment lifetime in years and M_{H_2} is the annual hydrogen production, as determined in equation (1), expressed here in kilograms of H₂ per year. This means that, depending on electrolyser utilization factor and total amount of hydrogen produced, the manufacturing emission contribution varies between projects.

Calculation of spatial variation in production emissions

The maps in Fig. 3 show the spatial variation in cradle-to-production gate GHG emissions of green hydrogen production for wind and solar photovoltaic electricity, using a polymer electrolyte membrane electrolyser. Sea water desalination is included to supply the water used for electrolysis (feed and cooling water) proportional to the local level of water stress²⁷ (landlocked countries excluded). For electrolyser-specific requirements and more details on water stress-dependent desalination calculations, see 'Electrolyser energy and material requirements' section. To calculate the local GHG intensities of wind and solar electricity, we combined the regression models of Dammeier et al.²⁴ (using solely wind speed and on- and offshore) and Bosmans et al.^{23,66} with wind speed and capacity factor maps from the GlobalWindAtlas (0.0025° resolution)⁶⁵ and solar irradiation and capacity factor maps from the GlobalSolarAtlas (0.0025° and 0.0083° resolution, respectively)⁶⁷. For offshore wind, we determine the footprint in the 'exclusive economic zone', which is the area up to 200 nautical sea miles from the coast. For solar electricity, we calculated the GHG footprint based on a solar photovoltaic facility for the year 2021 (Supplementary Information section 3), which we use to represent the most recent market, and resampled the local capacity factor map for solar photovoltaic to a

0.0025° resolution. For both wind and solar, the GHG footprint of green hydrogen production was calculated for three configurations: 'off-grid: curtailment', 'grid-connected: power export' and 'grid-connected: power import' (Table 1). For the 'grid-connected: power import' configuration, we used additional information on the national 2030 grid mixes modelled for a 2 °C policy scenario²⁶ to supplement the local renewable capacity factor to reach an electrolyser utilization factor of 1. For visualization purposes, all output maps were resampled to a 0.025° resolution by taking the mean value of all underlying cells. All analyses were performed in R (version 4.1.2), R Studio (version 2023.06.0) and ArcGIS Pro (version 3.1.2).

Emissions of alternative hydrogen production routes

The GHG footprint of grey hydrogen depends on the upstream methane leakage rate, ranging from 10.5 kgCO₂e kg H₂⁻¹ for a methane emission rate of 0.2% (lower bound) to 17.1 kgCO₂e kg H₂⁻¹ for a rate of 8% (upper bound)³⁴. A methane emission rate of 1.5% is considered a 'representative mid-range value'³⁴, resulting in a GHG footprint of grey hydrogen of 11.5 kgCO₂e kg H₂⁻¹, which we consider as the default value for grey hydrogen unless otherwise specified. The GHG footprint of blue hydrogen, assuming a CO₂ capture rate of 93%, is 2.3–8.9 kgCO₂e kg H₂⁻¹ for the methane emission range of 0.2–8% (ref. 34), calculated using electricity with a GHG intensity of 10 gCO₂e kWh⁻¹ (ref. 47). The GHG footprint of blue hydrogen is 3.3 kgCO₂e kg H₂⁻¹ for the mid-range value of 1.5% methane emission³⁴, which we consider as the default value for blue hydrogen.

Policy-based emission thresholds for clean hydrogen

A selection of policy-based GHG emissions thresholds for 'clean' or 'green' hydrogen is listed in Supplementary Table 10, for different system boundaries. The European Commission Taxonomy's threshold for sustainable hydrogen production of 3.0 kgCO₂e kg H₂⁻¹ is based on an emission reduction of 73.4% compared with grey hydrogen⁸³.

Data availability

All data used to produce the outputs presented in this paper can be accessed via Zenodo (<https://doi.org/10.5281/zenodo.11203454>)⁸⁵. We used publicly available data from the IEA Hydrogen Projects Database (version of October 2022) for hydrogen facility-specific information on electricity source, electrolyser technology, location and project size (accessible via <https://www.iea.org/data-andstatistics/data-product/hydrogen-projects-database>). For the calculation of location-specific emissions of solar electricity, we used the solar irradiance map available from Global Solar Atlas 2.0, a free, web-based application developed and operated by the company Solargis s.r.o. on behalf of the World Bank Group, utilizing Solargis data, with funding provided by the Energy Sector Management Assistance Program (ESMAP). For additional information, see <https://globalsolaratlas.info>. For the calculation of location-specific emissions of wind electricity, we used wind speed maps available from the Global Wind Atlas 3.0, a free, web-based application developed, owned and operated by the Technical University of Denmark (DTU). The Global Wind Atlas 3.0 is released in partnership with the World Bank Group, utilizing data provided by Vortex, using funding provided by the Energy Sector Management Assistance Program (ESMAP). For additional information, see <https://globalwindatlas.info>. To calculate the emissions of wind electricity based on wind speed and the onshore and offshore location, we created a generalized linear model based on wind turbine data from <https://doi.org/10.1111/jiec.13325> (ref. 24). We used the GHG intensities of national 2030 grid mixes modelled for a 2 °C policy scenario published by Knobloch et al.²⁶ at <https://doi.org/10.1038/s41893-020-0488-7>. For calculating sea water desalination requirements, we used publicly available data on country-level water stress scores from the World Resources Institute (<https://doi.org/10.46830/writn.18.00146>)²⁷.

Code availability

All code used to produce the outputs presented in this paper can be accessed via Zenodo (<https://doi.org/10.5281/zenodo.11203454>)⁸⁵.

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Acknowledgements

M.A.J.H. was financed by Grant 016.Vici.170.190 from the Netherlands Organisation for Scientific Research (NWO).

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K.d.K., S.V.H., M.A.J.H. and H.d.C. conceived and designed the study; K.d.K. performed the research; K.d.K. analysed the data with

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Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41560-024-01563-1>.

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Peer review information *Nature Energy* thanks Kevin Dillman, Petra Zapp and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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